

Investigating Daily Team Meetings in Agile Software Projects

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Abstract—An increasing amount of time is being spent at organizational meetings. One common type of meeting in software projects is the daily team meeting, which is the most important forum for coordinating and planning daily work. To better understand how software teams make decisions, communicate, and coordinate their work, we must uncover the micro-level interaction processes among the team members at these meetings. We analyzed transcriptions of eight daily meetings from two software development teams. The agile literature states that the daily meeting should focus on answering questions such as “What have I done? What will be done? What obstacles are in my way?” However, on average, only 24% of each of the meetings that we studied focused on this task. We found that 35% of the meeting was spent on elaborating problem issues and discussing possible solutions. Very little time was used for coordinating tasks. Our results indicate that many project decisions are made in daily team meetings and that this quick decision making requires team members to be experts. These experts need to have a shared understanding of who is responsible for what and of the information and requirements needed to solve the tasks.

Keywords: agile software development; daily meeting; stand-up meeting; Scrum; communication; coordination; decision-making; case study; software engineering

I. INTRODUCTION

Team meetings are necessary for coordinating team members, tasks and tools. Through meetings, a software team coordinates the efforts of those who are part of the development process and their activities and goals with suppliers, clients, and other groups both outside and inside the organization. Furthermore, meetings help the team solve problems, make decisions, exchange information, and strengthen team orientation. Team meetings shape both team and organizational outcomes [5]. Several studies have found that team performance is linked to the effectiveness of teamwork coordination [4, 7].

Recent development methods have argued that teams should conduct daily meetings [16, 20]. For example, agile software development (ASD) involves self-managing teams that rely on daily meetings to coordinate their work and make decisions. According to a recent survey, 78% of organizations that practice ASD conduct daily meetings [27]. The purpose of daily meetings in ASD is to

synchronize activities, improve communication within the team, and create work plans for the coming 24 hours [22]. Accordingly, in addition to being a decision-making forum, the daily meeting is the most important forum for exchanging information upon which decisions are based.

Daily meetings are becoming more common in software projects, and where these meetings are used, they are probably the most important forum for the team to coordinate and plan the daily work. Still, to our knowledge, no studies have examined such meetings. Therefore, to better understand how software teams make decisions, communicate, and coordinate their work, we need to acquire more knowledge on how team members interact in such meetings. This goal led us to the following research question: *how do team members interact with each other at daily meetings?*

To investigate this research question, we analyzed the transcripts of daily meetings at two companies. The aim is to understand how team members interact with each other, not to explore how team members' personalities or team leadership affects the daily meeting. The remainder of the paper is organized as follows. Section 2 outlines the background and relevant literature on daily meetings. Section 3 describes the research methodology. Section 4 reports on the results from our case study. Section 5 discusses our findings. Section 6 concludes and suggests future research.

II. BACKGROUND

Teamwork is critical for software companies, with meetings serving as an important means for sharing information and knowledge, coordinating activities, and combining team members' expertise. A software developer usually spends several hours a week in various types of meetings, including planning meetings, retrospective meetings, review meetings, customer meetings, and daily meetings. Because research on meetings in the field of computer science is lacking, the next section (A) covers the relevant research on meetings conducted within the fields of psychology, management, and small group research. Section (B) introduces the notion of daily meetings and research that may be relevant to such meetings.

A. Meetings

Rogelberg et al. [17] found that employees spend, on average, 5.6 hours in scheduled meetings during a typical work week. While meetings are necessary for building successful teamwork [5], several studies have reported that a great portion of time spent in meetings is perceived as ineffective [3, 12, 18].

Kauffeld and Lehmann-Willenbrock [5] analyzed micro-level processes in team meetings to better understand what makes these types of meetings successful. Their results showed that team meetings with more functional interaction, such as problem-solving interaction and action planning, were associated with higher team productivity [5]. In addition, they found that positive procedural communication in team meetings (i.e., statements aimed at structuring and organizing the discussion) was positively linked to team success.

Scholars have studied different meeting characteristics, such as the format and frequency of meetings. Bluedorn et al. [1] found that meetings where people stand (instead of sit) are shorter without lowering the quality of the decisions produced in the meeting. Luong and Rogelberg [9] found that frequent meetings, such as daily meetings, were potentially disruptive and that “the number of meetings an employee had per day was related to increased daily fatigue as well as greater subjective work load”. Additionally, Zijlstra et al. [31] found daily meetings to be problematic because “being interrupted several times has a greater effect than one interruption that takes longer”.

B. Daily meetings in agile software projects

The practice of conducting daily meetings in software development teams became popular with the introduction of agile methods, such as XP and Scrum, where daily meetings are a mandatory practice. Daily meetings are now common in agile software projects [27].

In the context of Scrum, the daily meeting is often called the “Daily Scrum” [22]. The term “Stand-up meeting” is also used to emphasize the fact that all participants should stand to avoid long discussions at the meeting [30]. The meeting is supposed to last no more than 15 minutes, and the purpose of the meeting is to improve communication, highlight and promote quick decision making, and identify and remove impediments [21]. According to Schwaber [22], the team members should discuss (1) what has been accomplished since the last meeting, (2) what is intended to be accomplished before the next meeting, and (3) which obstacles prevent the work items in the backlog from being accomplished.

While no studies focus specifically on daily meetings, we found three studies that investigated daily meetings as one of several agile practices. In the first study, Pikkarainen et al. found that daily meetings were a good way to keep the developers, project leader, and customers aware of the project status and that they helped developers resolve design issues more quickly [14]. McHugh et al. found that daily meetings had a positive

effect on communication inside a software development team [10]. Pries-Heje and Pries-Heje studied globally distributed teams and found that daily meetings are one of the four aspects of Scrum that provide coordination in particular and that the meetings allow all team members to interact and learn what the others are doing [15].

In our earlier research, we found that the daily meeting was seen as an important forum for information flow in the project [24] and to defend decisions and show other team members that a great deal of work has been accomplished since the previous meeting [25].

III. RESEARCH DESIGN AND METHODOLOGY

The background for this study was a large research program on agile teamwork, where the first two authors worked closely with several agile software companies.

Because the goal of this study was to explore and provide insight into daily meetings in agile software development, it was important to study software development teams in practice. According to Yin, case studies are the preferred research strategy if a “...question is being asked about a contemporary set of events over which the investigator has little or no control” [29]. In this paper, we reported on a single case study [29] where we studied one phenomenon in two projects at two different software companies.

When conducting the single case study, we followed the five-step process proposed by Yin:

1. Case study design: objectives are defined, and the case study is planned.
2. Preparation for data collection: the procedures and protocols for data collection are defined.
3. Collecting evidence: data on the studied case are collected.
4. Analysis: the collected data are analyzed.
5. Reporting: the results are reported.

A. Study Context

The researchers had a long relationship with the two SME companies studied in this paper, which we named TeleLink and MapTech. We chose these two companies because they were both part of a research project on agile teamwork and fulfilled the selection criteria that made it possible to answer the research question. In particular, they had implemented all of the Scrum practices, and we were able to observe some of their teams over a period of one year. In particular, we studied their daily, retrospective, review, and planning meetings.

Case studies can be based on any mix of quantitative and qualitative evidence, and having multiple sources of evidence ensures construct validity and achieves triangulation. In this study, we relied on the triangulation of our data sources, which is described in Section B.

1) TeleLink

TeleLink is an international software company that specializes in telecommunications. The project that we

studied aimed to develop a new system that would then be integrated with a larger software suite developed by the Finnish branch of the company. The project used Scrum with a sprint length of three weeks. The Product Owner of the project was located in Finland. The team consisted of eight people: a Scrum Master, four developers, two testers, and one technical document writer. At TeleLink, people worked regularly from home. Consequently, one or two team members attended the meetings by phone. The average number of years that the team members had been working at TeleLink was 13.3 years.

2) MapTech

MapTech is a company that develops software systems for archiving and systems for planning and coordinating excavation work. The Scrum project we studied produced a combination of textual user interfaces and map functionality. Six developers, one Scrum Master, and a Product Owner were allocated to the project. The Product Owner, who was also responsible for testing, was situated in another city and attended the meetings by phone. The Scrum Master was a part of both the quality and marketing departments and had worked in the company for more than 10 years. The length of each sprint was four weeks.

B. Study Design, Data Sources, and Analysis

We observed several daily meetings at TeleLink and MapTech and analyzed eight of them in detail. The meetings were supplemented by notes, interviews, observations from other meetings, and other materials collected from both organizations over a period of one year.

The meetings at MapTech were transcribed literally onto a computer during the observations by the second

author, who is an experienced typist. The TeleLink study started after the MapTech study was finished, and because some of the team members at this company spoke in a low voice, we had to tape and transcribe the observed meeting. Because we collected the same type of data from both companies, we considered all our data as part of the same case study. We coded the transcripts using descriptive coding [19]. Descriptive code uses a word or a short phrase to identify the *topic* (as opposed to the *content*) and is useful for answering questions such as “What is going on here?” Being with the teams for a year enabled us to understand the technical and business-related discussion, which is necessary for identifying a topic.

We conducted an extensive set of iterative analyses for each of the eight transcribed daily meetings. In the first iteration, the first two authors together coded each meeting manually on hard-copy printouts. During the next iterations, the transcripts were coded and recoded electronically with the help of a tool called NVivo (www.qsrinternational.com), which is used to analyze qualitative data.

The first two authors coded the micro-level processes by assigning a piece of text to a category (“node” in Nvivo). A category represents a topic on the content of the meeting. When categories were found to be conceptually similar in nature, we grouped them under more abstract, higher-order categories.

The categories evolved through discussions of the transcripts. At the same time, we reviewed the literature on design meetings [13] and a recently validated instrument for group interaction analysis [5]. We discussed each statement thoroughly before assigning it to a category. All authors participated in discussing and agreeing upon the final eight categories; see Table I.

We coded 100% of the words in the transcripts. That is, we coded all of the text and assigned each piece of text to

Table I: Coding scheme

No.	Category	Explanation	Examples
C1	The three Scrum questions	What have I done since the last meeting? What will be done before the next meeting? What obstacles are in the way?	Informing other members about the tasks the individual is working on.
C2	Problem-focused communication	The major questions and problems that need to be addressed, including the elaboration of the issue and discussions of possible solutions to the problems.	Discussing questions such as “How can we implement that feature? How do we integrate the components? What is the best solution to the identified problem?”
C3	Clarification	Explanations that make an earlier event, situation, or statement clear.	Questions that someone asked to better understand an issue.
C4	Criterion	The reasons or arguments that evaluate an alternative solution or proposal.	Arguments for a solution, customer requirements, and technical possibilities.
C5	Coordination of tasks	Delegation of tasks and assignment of responsibilities.	Discussing who should be involved in solving a task.
C6	Meeting management	Statements related to the orchestration of the meeting activity.	Indicating that members hold off on discussions, asking someone to speak, and summing up the meeting.
C7	Project management	Statements concerning activities not directly related to the content of the meeting.	Discussing the reporting of resources and hours and providing information about other meetings.
C8	Digression	Discussions of side topics or interruptions related to things outside the content of the meeting.	Telling a joke, interrupting because of technical problems with phone conference equipment, and other distractions.

only one of the eight exclusive categories.

When the coding of the transcripts was done, we identified the transitions and sequences among the categories to determine what type of categories led to what type of interaction.

To identify the proportion of each category at a meeting, we used the number of words spent on a category and then calculated the percentages of the occurrence of different categories at the meetings. We believe that this classification is a valid method for identifying the proportion of each category because people were talking for most of the time at these meetings. We analyzed a total of 11,237 words from the eight meetings.

IV. RESULTS

We now report the results from analyzing the data on the team interactions during their regular daily meetings at MapTech and TeleLink.

A. Meeting Characteristics

Table II summarizes the characteristics of the observed meetings.

Table II: Meeting characteristics

Company	Length in minutes	Total # of people in meeting	# on phone
MapTech	19	7	1
MapTech	15	7	1
MapTech	20	7	1
MapTech	12	5	0
TeleLink	14	6	0
TeleLink	20	5	2
TeleLink	21	8	2
TeleLink	13	8	1
Avg:	16.8	6.6	1.0

The length of the daily meetings varied from 12 to 21 minutes depending on the number of people who attended the meetings, the technical problems at the meetings, and most importantly, the amount of time spent on discussions related to *problem-focused communication* (C2).

At three of the eight meetings, the Scrum Master announced that the daily meeting was officially over, but the meeting still continued, with the Scrum Master giving general information about the project or team members discussing among themselves. Thus, in reality, these meetings lasted longer than the time periods shown in the table. We also observed that people often arrived late for the meetings. As a result, the meetings seldom started on time.

B. Percentage of Words Devoted to Each Category

The percentage of words devoted to each of the eight categories is shown in Figure 1. The two teams across the two organizations were remarkably similar in terms of the proportion of words they devoted to each category.

The largest category at the meetings was *problem-focused communication* (C2, Table I), accounting for 35% of the statements at the meetings. The discussions of problems and solutions usually involved several team members. We found that many decisions were made during these discussions.

The three Scrum questions (C1), the second largest category, constituted, on average, 24% of the statements at the meetings. In this category, 53% of the statements described *what was done since the last meeting*, 25% covered *what will be done*, and 22% covered *obstacles*.

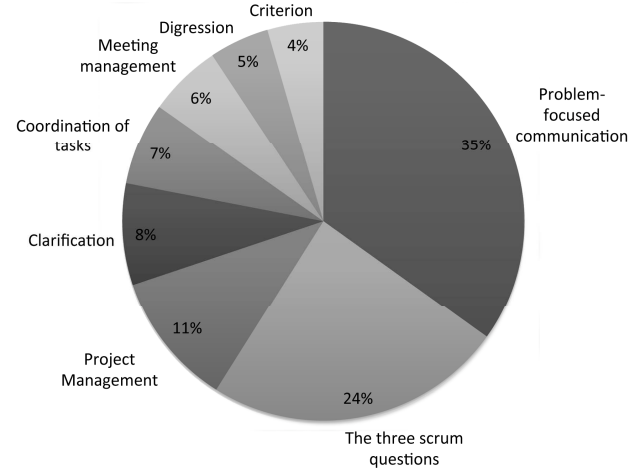


Figure 1. Proportion of each category in the meetings

Project management (C7), which constituted 11% of the meetings, included discussions on the number of hours that had been spent on specific tasks, the number of hours that were left on a task, and the process of reporting these hours in the time reporting system.

The teams spent 8% of the meetings on *clarification* (C3). Statements in this category mostly requested more information or asked a team member to explain their previous statement. There were, on average, six clarifications at each meeting, with three to nine clarifications per meeting.

Coordination of tasks (C5) constituted 7% of the meetings. Both the Scrum Master and the team members were involved in coordinating tasks. Sometimes, tasks were directly assigned to a team member, and at other times, team members voluntarily took on responsibilities.

The teams spent 6% of the daily team meetings on *meeting management* (C6). One example of meeting management is when the Scrum Master asks the team members to speak or to hold off on long discussions. Another example is when the Scrum Master leads the discussion back to the topic.

Digression (C8) constituted 5% of the meetings. We coded three types of digressions: (1) discussions of side topics; (2) trouble with meeting equipment, such as Wi-Fi or telephones; and (3) statements that lightened the atmosphere and often resulted in laughter at the meeting. Among the digressions, 40% lightened the atmosphere,

32% were side topics, and 28% were statements about trouble with equipment.

Criterion (C4), which mostly concerned customer requirements, accounted for 4% of the statements made at the daily meetings. Two of the meetings we studied had no statements in this category.

C. Sequences of Categories

We analyzed the interaction processes at the meetings and identified the various patterns of a sequence of interaction. While several patterns were identified, the pattern depicted in Figure 2 was the most common sequence of interaction at the meetings and was also the pattern that included all three Scrum questions (C1).

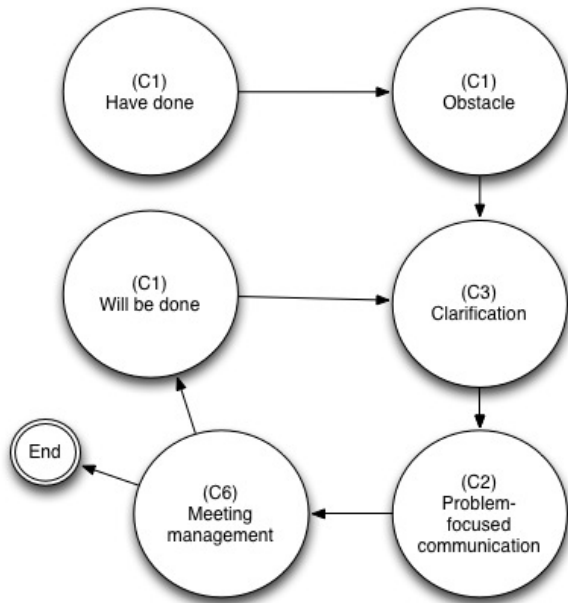


Figure 2. The most common sequence of interaction in a daily meeting

When following this pattern, a team member usually started by telling the team what he or she had done since the last meeting (C1) before continuing to discuss obstacles (C1). An obstacle often caused another team member to ask for clarification (C3) regarding the obstacle just explained. The resulting clarification often led to a discussion of the identified problem. When the discussion ended, the Scrum Master usually asked the team member to talk about what he or she planned to do (C1). We found that by describing what he or she planned to do, the person often triggered a new clarification, which again led to a new discussion about problems and solutions. This discussion often ended with the Scrum Master summing up the discussion and asking the next person to describe what he or she had done since the last meeting.

Problem-focused communication (C2) was the category that accounted for the largest proportion of statements at all of the daily meetings. To better understand the daily meetings, one must understand how *problem-focused communication* was initiated at the

meetings and allowed team members to discuss their major issues, problems, and solutions. We analyzed the transitions to problem-focused communication. The total number of transitions leading to this category was 46. Among the 46 transitions observed, the three most common transitions were *clarifications (C3)*, *obstacles (C1)*, and *coordination of tasks (C5)* (Figure 3). Asking for a clarification was the most common reason why the team would end up in a problem-focused discussion. In the second most common transition, a team member would report on an obstacle, and the team or part of the team would then promptly commence a discussion of the problems (*problem-focused communication*). The third most common transition to *problem-focused communication* occurred when the team discussed how to coordinate tasks. When the team was discussing who should be engaged in solving a task, people clearly became aware of new challenges and problems.

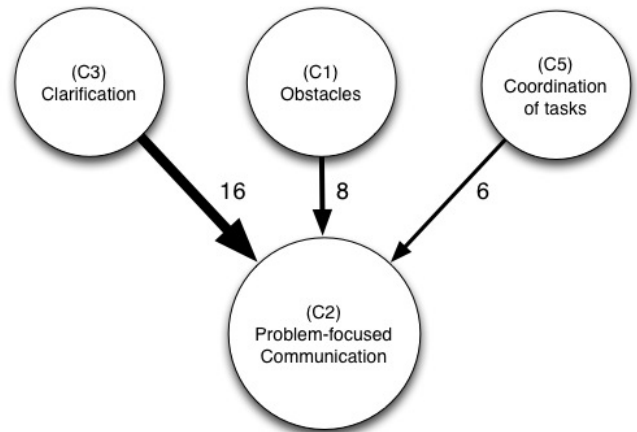


Figure 3. Transitions to problem-focused communication

V. DISCUSSION

While the team meeting was intended to focus on answering *the three Scrum questions*, we found that most of the meeting was spent on other types of interactions instead. During the meeting, the team members talked most often about *problem-focused communication*. *Problem-focused communication* included 46% more words than the second biggest category, which comprised *the three Scrum questions*. Further, we found typical patterns of interaction processes and transitions at the meetings. Our data also suggest that the discussions at daily meetings were similar across different organizations, both in terms of the number of words spent on a category and the sequential relations among the categories.

In this section, we discuss our key observations in light of our research question: *how do team members interact with each other at daily meetings?*

A. Interaction categories and processes at daily meetings

The literature has usually described daily meetings as a place for answering *the three Scrum questions*. However,

we found that this category only accounted for 24% of the statements at the meeting. From this point, we understand that, while the meeting was said to focus on answering *the three Scrum questions*, most of the meeting was spent on other types of discussion.

Because we found that the teams spent most of the meeting on *problem-focused communication*, we argue that the daily meeting was mostly about making operational decisions. Although the number of issues and discussions of the major problems and solutions differed across the eight meetings, the average number of issues was similar for both organizations. The next section further describes the role of decision making at daily meetings.

The percentage of statements at meetings spent on the *coordination of tasks* was surprisingly small (7%) given that one ostensible goal of the daily meetings was to coordinate activities for the next 24 hours. This finding indicates that daily meetings in agile software projects are not the primary meetings for coordinating tasks. Another explanation is that coordination is simple and requires little effort in such meetings, especially if the teams have been working together for a long time, which was the case in our study.

Clarification was the main category that initiated *problem-focused communication*. This finding demonstrates that clarification plays an important role in initiating problem solving. Our findings are in accordance with Olson et al. [13], who found that spending time on clarification was an important characteristic of collaborative design meetings. In the meetings we studied, *clarifications* constituted 8% of the statements made at the meetings. One reason for this relatively small figure was that clarification often consisted of relatively short questions, such as “Can you explain more?”

We found that *meeting management* was important for structuring communication at the meetings. We observed that several longer discussions were cut short by the Scrum Master when he asked the team members to hold off on long discussions. We also observed that the Scrum Master frequently asked someone to speak. We believe that these requests made the Scrum Master more of a leader than a facilitator of the meeting and that it contradicted the self-organizing nature the meeting could have had if the team members themselves had decided that the next person could speak.

Only 5% of the meetings were spent on *digressions*. The largest group of digressions was “Lightening the atmosphere”, which is often referred to as “Positive socio-emotional communication” [5]. Kauffeld and Lehmann-Willenbrock, who studied teams that have worked together for some time, found that positive socio-emotional communication was not positively connected to team meeting success [5].

When a person was requested to speak, the interactions among the team members often followed the common pattern shown in Figure 2. Our analysis showed that team members often discussed obstacles that they faced before they talked about what they were going to do. In contrast,

the Scrum literature indicates that the reverse should occur. Apparently, the team members perceived it more important to discuss obstacles than future work. Additionally, by addressing problems, the members could explain why they had not followed through on the previous day’s plans (i.e., the person felt the need to justify why he or she was having problems with progress). This technique is also known as self-justification [23].

B. Decision making in daily meetings

A successful problem-solving process is often characterized by a thorough definition and analysis of the problem [28], which can be time consuming. However, from our analysis, we found that most decision-making sessions could be as short as seconds or minutes. Daily meetings are characterized by limited time, uncertainty, and incomplete information. In such a context, it is difficult to discuss several solutions to a problem before making a decision.

To understand the effect of decision making in a complex environment with time pressure, we use the theory of naturalistic decision making (NDM) [6, 11]. NDM describes how people use their experience to make decisions in complex, dynamic, real-time environments [11]. NDM postulates that experts can make good decisions under difficult conditions, such as time pressure, uncertainty, and vague goals, without having to perform extensive analyses and compare options. They are able to do so by employing their experience to recognize problems that they have previously encountered and for which they already developed solutions. Experts use their experience to form mental simulations of the problem currently being encountered and use these simulations to suggest appropriate solutions [11]. Accordingly, we argue that agile teams apply naturalistic decision making when conducting daily meetings.

There are certain implications of viewing daily meetings as an NDM process. First, companies need to train their employees to develop domain-specific expertise and collaborative teamwork skills. Second, because NDM relies on highly trained experts, agile teams need to consist of experts, not novices. If there are novices in the team, companies need to determine how to move the novices through the stages until they become experts. Third, to make teams perform effective decision-making processes at such meetings, the team members need to have shared mental models; that is, they must have a shared understanding of who is responsible for what and of the information and requirements needed to solve the tasks [8].

While using the daily meeting for frequent decision making under time pressure presents some challenges, problem-focused communication and knowledge sharing at such meetings are positively linked to team success [5]. The same is true of action planning, such as coordination of tasks. The three categories *clarification*, *criterion*, and *meeting management* are examples of what is called “positive procedural communication” and are positively linked to team meeting success [5]. Consequently, the

daily meeting is important because it forces the team to spend time on interaction processes and categories that would support their success.

Although spending time on solving and discussing problems is good for the team, one challenge might be that decisions are not documented. The teams that we observed did not document the decisions taken in the daily meetings. Because some decisions might be wrong and have major consequences, they should be recorded and re-evaluated. In addition, we observed that people were sometimes absent from the meetings and thus missed important information because there were no minutes. Moreover, we did not observe team members updating each other on the outcome of the meeting.

One explanation for the lack of documentation at these meetings could be the dominant “lean” mentality that has been advocated in agile development. This mentality aims to minimize unnecessary work and to avoid wasteful documentation in particular. While this concept has been misconstrued by many to mean “no documentation”, the idea is that the team should document what is absolutely necessary and nothing more [2]. None of the teams discussed the need for writing minutes.

VI. LIMITATIONS

One limitation of our study is that the presence of a researcher might have influenced the participants’ mental processes in the discussions. However, because we visited and observed the teams over a long period of time, the participants became used to being observed, and we believe that the meetings would have been the same without the researchers being there.

In one company, we transcribed the meetings while they were happening. We realize that doing so might have influenced our results because the researcher might not have captured all of the words. However, if a few words were lost in this process, the loss would likely be equal for all categories and topics. Therefore, the omissions would not affect our results. The length of meetings listed in table II (Meeting characteristics) was not affected by these potential omissions.

We studied only eight meetings from the two projects in the two companies. These meetings may not have been representative. However, by being in the company for a long period of time and observing several meetings, we believe we were able to pick typical meetings for data analysis.

VII. CONCLUSION AND FURTHER WORK

This paper reported data from a case study with the aim of uncovering the categories of communication at daily meetings to better understand how team members interact at these meetings. Such knowledge is important for understanding the role of daily meetings in particular and teamwork in software projects in general.

Most agile software development projects conduct daily meetings where each team member must answer *the three Scrum questions*: “What have I done? What will be

done? What obstacles are in my way?” We therefore anticipated that these questions would account for the largest proportion of statements at all of the daily meetings. However, this proportion comprised only 24% of the statements. The largest category was *problem-focused communication*, which focuses on understanding issues and problems and discussing solutions. This type of communication allows many decisions to be taken at the meeting. Further, we can understand the decision-making process at the daily meeting by using the theory of naturalistic decision making, which has implications for who is on the team and how a company organizes its training.

Thus, returning to the question: *how do team members interact with each other at daily meetings?*, we found the following:

- The daily meeting mostly centered on *problem-focused communication*.
- Discussing *obstacles* and *what will be done* triggered *problem-focused communication*.
- The team performed a lot of quick decision making under time pressure without considering alternatives, which can be understood as naturalistic decision making.
- The Scrum master usually asked someone to talk and sometimes had to lead back to the topic.
- The team did not spend much time on coordinating tasks.

The implications of these findings are as follows:

- The team must consist of highly trained experts with a shared understanding of tasks and goals.
- Discussing obstacles and what will be done (Questions 2 and 3 in Scrum) seems to be most important because it triggers problem-focused communication.
- To strengthen the self-managing agile team, the Scrum master should not ask people to speak.
- The team needs to identify how to document and follow up on important decisions.

Our findings are based on only two software projects. Future research should analyze the interaction processes in other projects. In addition, the efficiency of meetings should be measured by interviews and questionnaires. More research is required to examine the relationship between interaction processes in meetings and long-term effects, such as the effects on team performance and organizational success.

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